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Evaluating Tools for Automatic Reconstruction of Clonal Evolution

Background

In the field of cancer research, analyzing the clonal evolution of a tumor is of major interest. The early identification of aggressive subclones can enable new therapeutic options. Therapy resistance might be detected prior to relapse, thus, having the potential to influence survival significantly [1]. However, the actual reconstruction of clonal evolution on the basis of mutational data is not trivial. Despite numerous algorithms existing, many publications reporting on clonal evolution still choose manual reconstruction as the most reliable method [1, 2].

Methods

We performed extensive research on currently available tools for the reconstruction of clonal evolution considering bulk mutational data. Tools were split into three categories: 1) clustering variants, 2) constructing the clonal evolution tree, 3) visualizing clonal evolution. All tools meeting at least one of these three categories were taken into account. Performance of the tools, focusing on each tools' main functionality, was investigated using two minimum examples of clonal evolution, consisting of two time points: Seven mutations with distinct allelic frequencies were supposed to form five clusters. Example 1 considers linear clonal evolution, while example 2 considers branched clonal evolution.

Results

Clustering: Altogether, we identified 7 tools performing clustering of variants. All tools could be installed, executed and reported results on our exemplary data. However, only 1 tool (PyClone) reported the expected number of five clusters (see Figure 1).

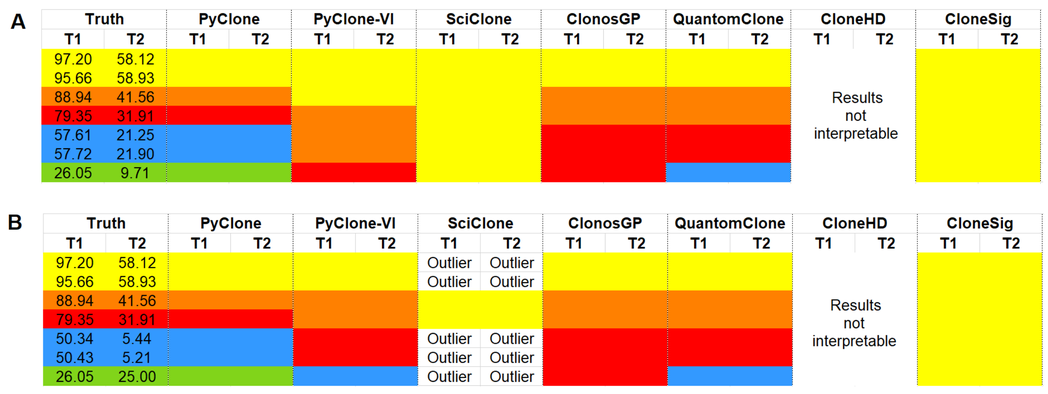

Figure 1: Clustering of seven exemplary mutations at two time points. A) Linear clonal evolution; B) branched clonal evolution.

Tree reconstruction: Ten tools performing tree reconstruction were identified. Using the correct clustering output of the previous step as input, we evaluated performance of the algorithms. SiCloneFit had to be excluded from evaluation as it exclusively works on single-cell DNAseq data. CITUP, Canopy and ClonEvol reported errors for both minimum examples, not generating any output. AncesTree could not identify a solution for any of our minimum examples. PhyloWGS reported 'trees' with only one node containing all variants for both examples. PASTRI performed considerable adjustment of cell fractions, reported several trees and added a new clone containing undefined mutations. Altogether, only 3 tools, ClonalTREE, SPRUCE and SCHISM, reconstructed proper clonal evolution trees for both of our two minimum examples (see Figure 2). ClonalTREE does not consider clustering, but interprets every mutation as an individual clone. As a result, the algorithm did not succeed in reconstructing the correct phylogenetic tree for any of our two examples. SPRUCE reported the correct clonal evolution for one, SCHISM for both of our examples.

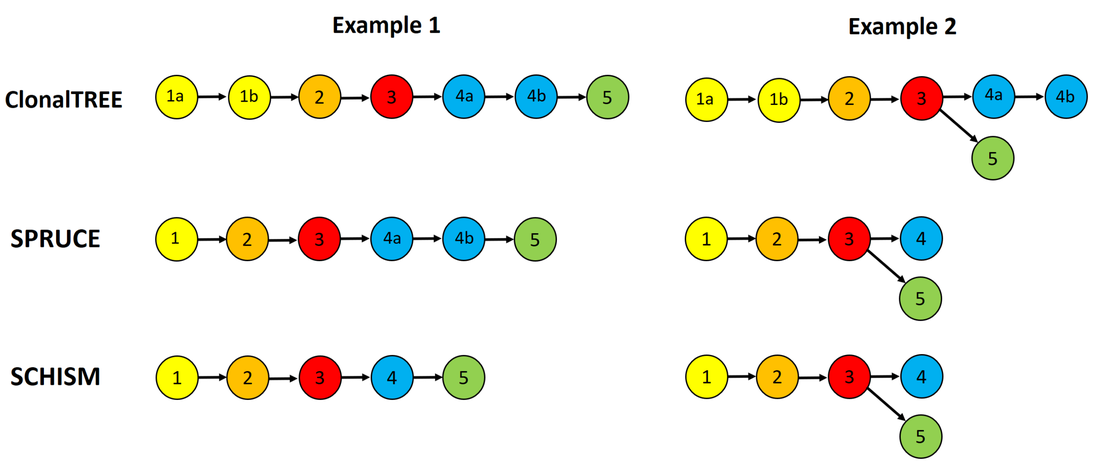

Figure 2: Phylogenetic trees reconstructed for two minimum examples using ClonalTREE, SPRUCE and SCHISM.

Visualization: For visualization we evaluated two approaches: fishplot and TimeScape. Both tools could be installed successfully and clonal evolution could be visualized for both examples (see Figure 3).

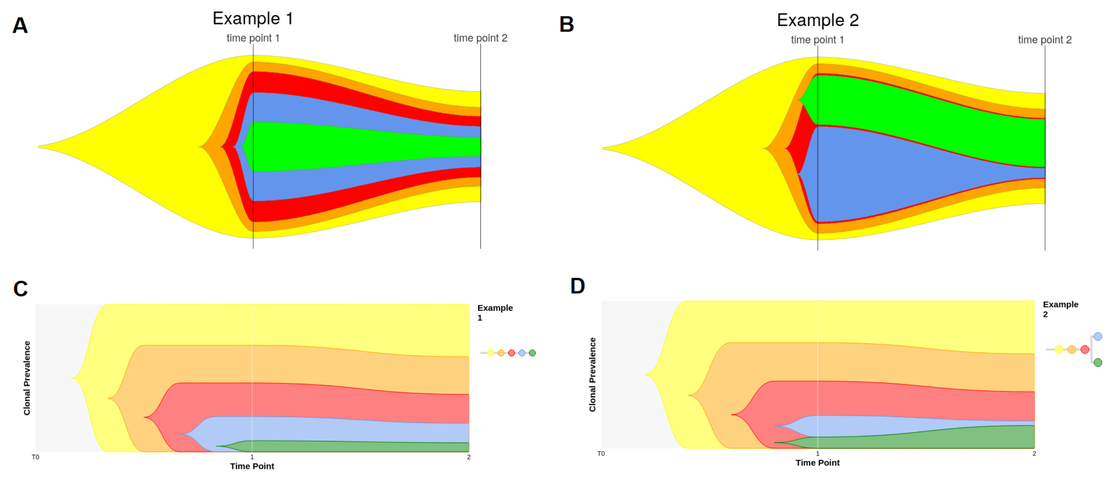

Figure 3: Visualization of two minimum examples of clonal evolution. A) Linear evolution example with fishplot. B) Branched evolution example with fishplot. C) Linear evolution example with TimeScape. D) Branched evolution example with TimeScape.

For fishplot it can be observed that clones 2 to 5 appear within short time. This leads to problems with respect to space in case of example 2. Clone 5 (green) appears to result from clone 2 (orange), although it originally evolves from clone 3 (red). If artificial time points, prior to the first actual time point are added, this problem can be resolved and clonal evolution can be visualized in a straightened way.

With regards to TimeScape it can be observed that the tool does not scale correctly considering the fraction of mutated cells. At time point 1, 97% of the cells are characterized by mutations. However, at time point 2 a decreased number of only 58% carries mutations in both examples. In Figure 3 C and D it can be observed that TimeScape displays the overall mutation load at both time points as 100%. Instead, fisplot displays the correct overall percentage of cells.

Discussion

Our evaluation of tools for automatic reconstruction of clonal evolution shows that despite many tools existing, only very few succeed in reconstructing the correct clonal evolution on the basis of our two minimum examples: for clustering, our results suggest that PyClone is still the best approach – even compared to its successor PyClone-VI. For the actual reconstruction of phylogenetic trees, SPRUCE and SCHISM performed best on our exemplary data. For visualization, we suggest to use fishplot, adding artificial time points. However, it remains to be analyzed how these approaches perform on real, more complex data, including a mixture of SNVs and CNVs as well as false positive variant calls.

In general, our observations indicate that manual reconstruction of clonal evolution is currently still greatly irreplaceable and there is a pressing need for better algorithms. Yet, the analysis of single cell data might be an option to approach the problem of automatic reconstruction of clonal evolution from a different angle.

References

[1] Reutter, K. *et al.* (2021) Reconstructing clonal evolution in relapsed and non-relapsed Burkitt lymphoma. *Leukemia*, **35**, 2; doi: 10.1038/s41375-020-0862-5. **[2]** da Silva-Coelho, P. *et al.* (2017) Clonal evolution in myelodysplastic syndromes. *Nat Commun*, **8**, 15099; doi: 10.1038/ncomms15099.

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